

Research Methodology: Class 9-10



Class-9: Sampling Considerations

Class-10: Data Processing

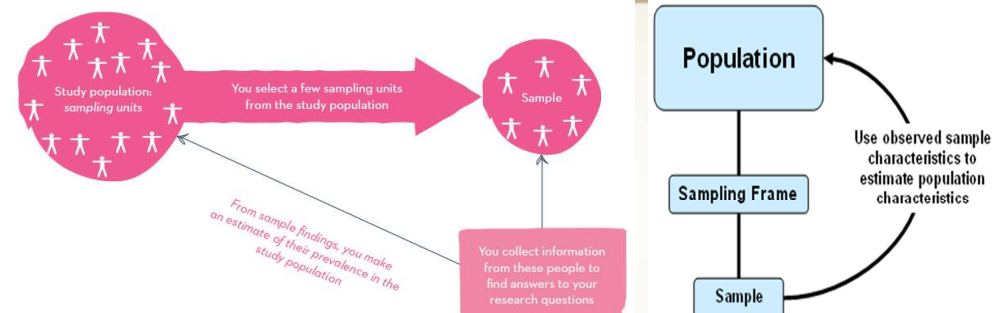
Learning Objective

By the end of the chapter, students should be able to answer these questions.

- What is the concept of sampling and how sample differ from the census?
- What is sampling and non-sampling error?
- Define the probability and non-probability sampling?
- What are the various types of probability and non-probability sampling?
- What are the primary factors affecting the calculation of sampling size for a population?
- How to calculate the sample size?

Sampling, population,
sample, census, sampling
frame

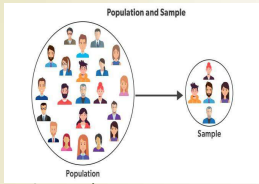
The concept of Sampling



Sampling: It is a process of selecting an adequate number of elements from the population so that the study of the sample results provides information about the characteristics of the population. In other words, it is the process of obtaining information about an entire population by examining only a part of it.

Population, Sample

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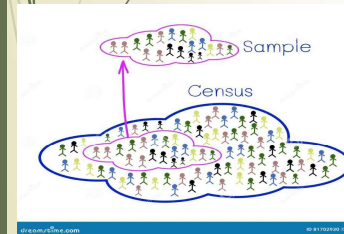
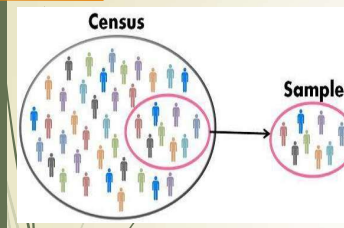


- A **population** is the collection of all the elements of interest. Example: The number of full-time MBA students in a business school
- The **sampling frame** comprises all the elements of the population with **proper identification** that is available to us for selection at any stage of sampling. Example: **Registered voters, Number of students in a university, Telephone Directory.**
- Sample Unit or sampling element:** A single member of a particular sample is referred to as a sampling unit.

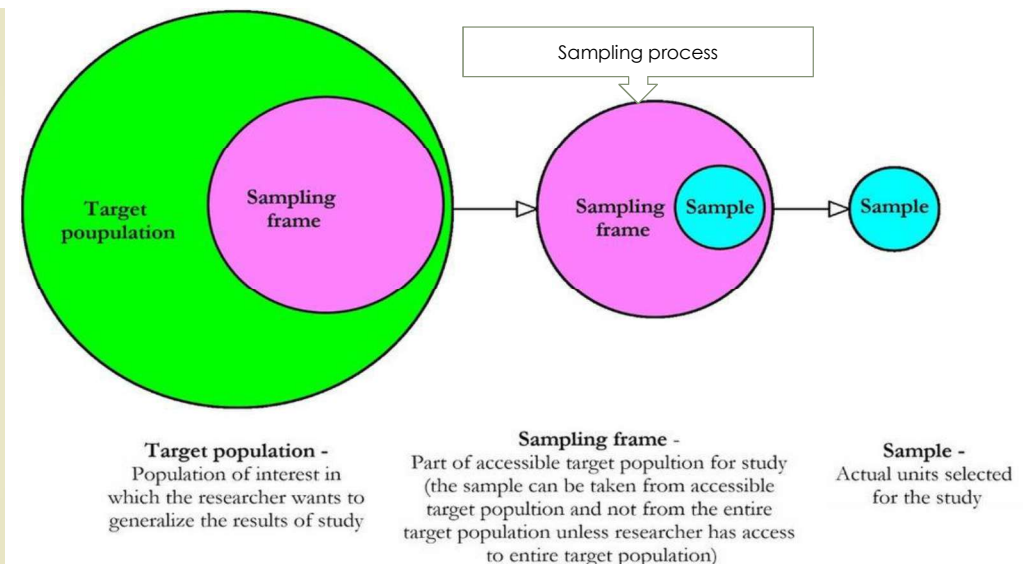
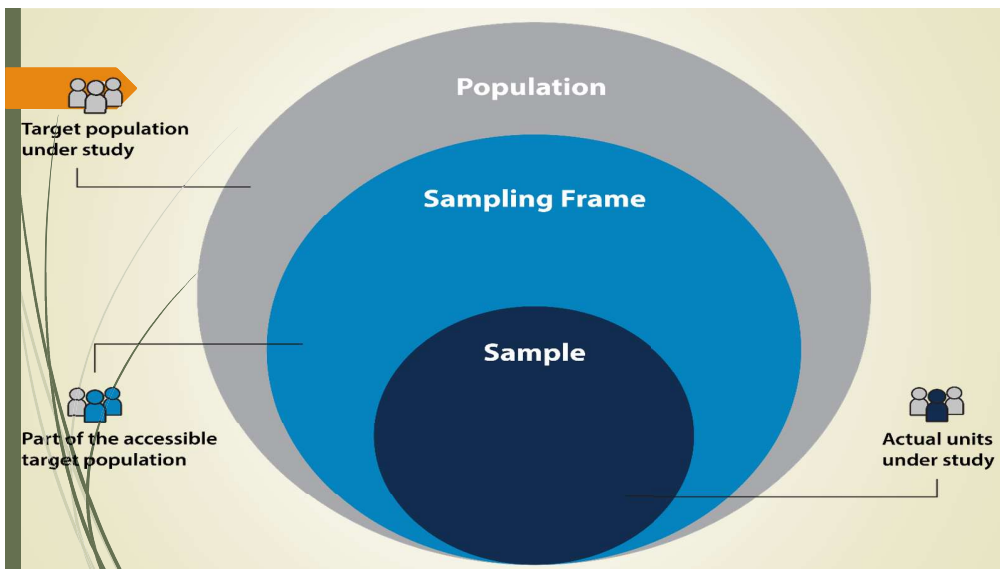


Population, Sample

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- A **sample** is a subset of the population. For example, 50 students are considered among 200 MBA students.
- Census:** A census is an examination of every element of the population with certain identity.
- Sample size:** The number of individuals from whom we obtain the required information.



Case-study-9.1: Population and Sample

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- A teacher, Mrs. Sharma, wants to know the average height of her school's 1,000 students.
- If Mrs. Sharma measures the height of **all 1,000 students**, she is collecting data from the **population**, that is, **every member** of the group of interest.
- But if she only measures the height of **100 students chosen randomly**, that smaller group is called a **sample**, it is a **subset** of the population, used to **estimate**.

Key Difference:

- ✓ **Population** = Entire group (all students)
- ✓ **Sample** = Part of the group (some selected students)

Case-study-9.2: Sample and Census

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- Imagine a farmer, Mr. Patel, who has **10,000 mango trees** in his orchard. He wants to know how sweet the mangoes are this season.
- ✓ If Mr. Patel **plucks and tastes mangoes from every tree**, he is conducting a **census**, collecting data from **every unit** in the population.
- ✓ If he **tastes mangoes from only 500 randomly selected trees**, he is using a **sample**, collecting data from a **small part** of the population to estimate the sweetness for all.

Key Difference:

- ✓ **Census** = Study of **every** member of the population
- ✓ **Sample** = Study of **only a few** members of the population

Case-study-9.3: Population, Sample, and Census

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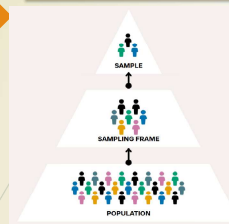
Story Theme: A Village Health Survey

Concept	Story Element	Explanation
Population	A village has 1,000 residents. The health department wants to know the average age of all residents.	This represents the entire group we want information about.
Census	The health workers go door-to-door and record the age of all 1,000 people .	This is a census , where data is collected from every individual .
Samples	Instead of visiting everyone, the health workers randomly choose 100 people and record their age.	This is a sample , a small group used to estimate the age of the whole village.

- **The population is the whole village.**
- **The census checks everyone in the village.**
- **The sample checks just a few people to represent the whole.**

Sampling frame

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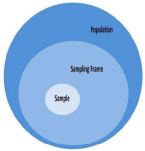


- In research methodology, a sampling frame is defined as: **A list or database of all the elements in the population from which a sample will be drawn.**
- It serves as the source from which the sample units are selected and is crucial **for ensuring that the sample is representative of the population.**

Study objective	Population	Sampling frame
Survey on student stress in a university	All students in the university	Official university enrolment list
Market study on dairy consumers in a city	All households in the city	City's voter list or utility customer database
Health study on elderly in rural areas	Elderly population in villages	Village census records or medical centre databases

Case-study-9.4: Sampling frame

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- **Research Study:** Investigating the prevalence of stress among high school students in a specific city and its impact on academic performance.
- **Population:** All high school students in a particular city
- **Sampling Frame:** The sampling frame involves obtaining a comprehensive list of all registered high schools in the specific city. A random selection of schools would be made from this list to ensure representation from different areas and demographics of the city.
- **Sample:** Randomly selected 500 high school students from different schools in the city
- The sample represents a subset of the entire population of high school students in the city.

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Sampling and Non-sampling error

Sampling and Non-sampling Errors

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Sampling Error: Sampling error occurs when the sample selected from a population does not perfectly represent the whole population.

In simple words: You study only a sample, so your result may be slightly different from the actual population result.

Example: Suppose a college has 1,000 students, and you select 100 students to find the average height.

- Actual average height of all 1,000 students = 165 cm
- Average height of the selected 100 students = 167 cm

The difference (2 cm) is called **sampling error**.

Key point: Sampling error happens because we study a sample instead of the entire population.

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Reasons for sampling error

- **Small Sample Size:** Example-A college has 5,000 students, but you survey only 20 students to find their average monthly spending.
- **Unrepresentative Sample:** Example-To know the average income of a city, you select only people from a high-income area.
- **Wrong Sampling Method:** Example-Selecting only people who are easily available instead of randomly selecting people from the entire population.
- **Variation Within the Population:** Example-If a city has both very low-income and very high-income families, different samples may produce very different average incomes.
- **Chance of Selection:** Example-One random sample may contain more young people, while another may contain more older people.

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Sampling and Non-sampling error

Non-sampling Error: Non-sampling error is an error that occurs **during the collection, recording, processing, or analysis of data**, and it can happen even if we study the entire population.

In simple words: **It is an error caused by problems in the way data is collected or handled, not because of selecting a sample.**

Example: Suppose you conduct a survey of **1,000 students** about their monthly expenses.

A student actually spends **₹10,000**, but the interviewer accidentally records it as **₹1,000**.

The wrong result is due to a **data recording error**, so it is a **non-sampling error**.

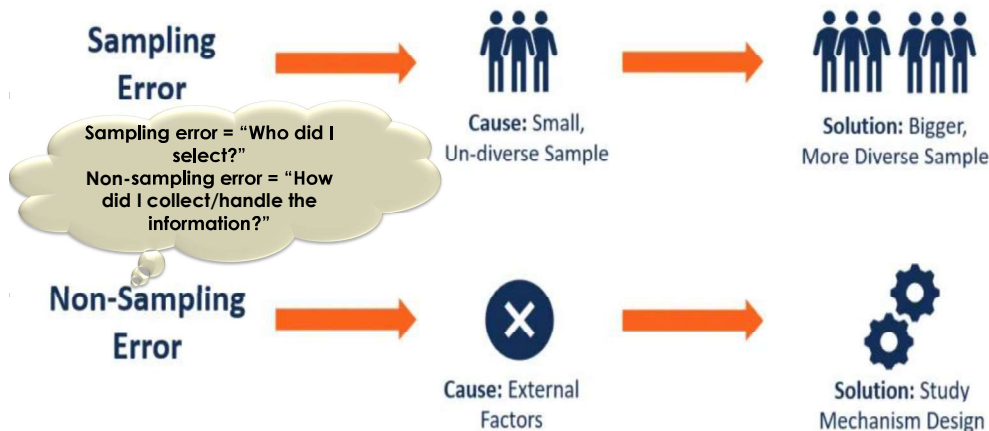
Key point: Non-sampling error occurs due to **mistakes in data collection, recording, processing, non-response, etc.**

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Reasons for non-sampling error

- **Non-response Error:** Example: Out of 100 people surveyed, 20 do not answer the questions.
- **Response Error:** Example: A person earns ₹30,000 but reports their income as ₹25,000.
- **Interviewer Error:** Example: The respondent says "Yes," but the interviewer records "No."
- **Questionnaire Error:** Example: Asking, "Do you often exercise?" without defining what "often" means.
- **Processing Error:** Example: ₹5,000 is entered into a computer as ₹50,000.
- **Measurement Error:** Example: A faulty weighing machine shows 70 kg when the actual weight is 65 kg.
- **Coverage Error:** Example: A survey of internet users uses a list that excludes people who access the internet only through mobile devices.

Reducing Sampling & Non Sampling Errors



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Case-study-9.7: : The School Election Survey

- Mrs. Rao, a teacher, wanted to predict who would win the school captain election. There were **500 students**, but she asked only 50 students, a sample to save time.

Sampling Error: She **randomly picked 50 students from Class 12**. Most of them supported **Ananya**. So, Mrs. Rao predicted that Ananya would win. But on election day, **Ravi** won. Why? Because the sample only included Class 12 students, who didn't represent the full school, especially the junior classes who mostly supported Ravi. **This is a sampling error**, caused because the **sample didn't represent the entire population** well, even though it was selected fairly.

Non-Sampling Error: In another case, Mrs. Rao asked students from different classes, but:

- Some students were **too shy to answer honestly**.
- One student **ticked the wrong name by mistake**.
- Another student **misread the question** and picked the wrong option.

As a result, the prediction was still wrong, **even though the sample was good. This is a non-sampling error**, caused by **mistakes in data collection, understanding, or response**, not the sample itself.

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Sampling Design

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Probability sampling

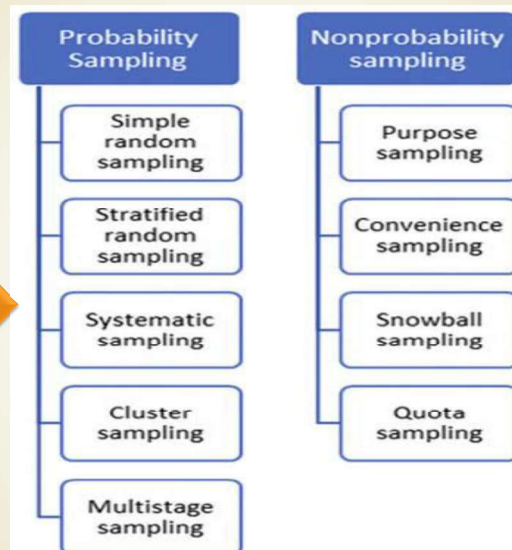
- **Probability sampling** is a technique in which researchers choose samples from a larger population based on the theory of probability. This sampling method considers every member of the population and forms samples based on a fixed process.
- **Probability sampling, also known as random sampling**, is a kind of sample selection where randomisation is used instead of deliberate choice.
- **For example**, in a population of 1000 members, every member will have a 1/1000 chance of being selected to be a part of a sample. Probability sampling eliminates sampling bias in the population and allows all members to be included in the sample.

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Non-probability Sampling

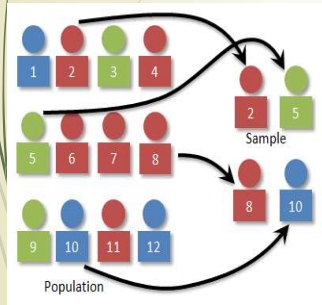
- **Non-probability sampling** techniques are where the researcher deliberately picks items or individuals for the sample based on their research goals or knowledge.
- **It does not rely on randomization.** This technique is more reliant on the researcher's ability to select elements for a sample. Outcome of sampling might be biased and makes difficult for all the elements of population to be part of the sample equally. This type of sampling is also known as non-random sampling.

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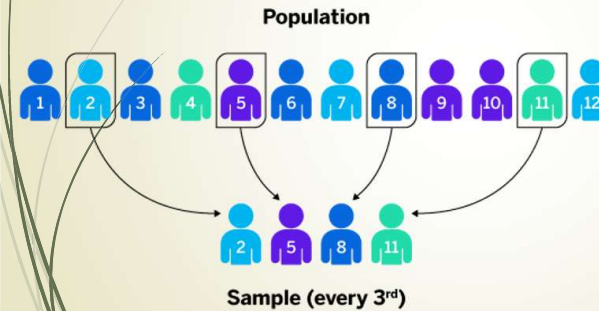
Probability sampling: Simple random sampling



- Every element has an equal chance of getting **selected** to be the part sample. It is used when we don't have any kind of prior information about the target population.
- For example: Random selection of 20 students from class of 50 student. Each student has equal chance of getting selected. Here probability of selection is $1/50$

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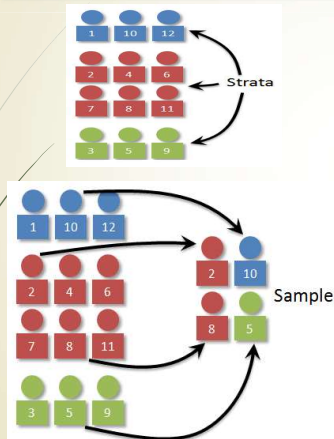
Probability sampling: Systematic sampling



- Here the selection of elements is **systematic** and not random except the first element. Elements of a sample are chosen at **regular intervals of population**. All the elements are put together in a sequence first where each element has the equal chance of being selected.
- Taking an example of $N=20$, $n=5$ No of elements in each of the subgroups is N/n i.e. $20/5 = 4 = k$
- Now, randomly select first element from the first subgroup. If we select $n_1 = 3$; $n_2 = n_1 + k = 3 + 4 = 7$; $n_3 = n_2 + k = 7 + 4 = 11$

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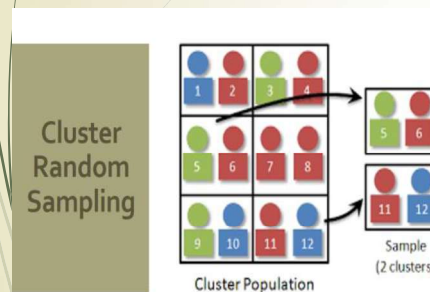
Probability sampling: Stratified Sampling



- This technique divides the elements of the population into small subgroups (strata) based on the similarity in such a way that the elements within the group are homogeneous and heterogeneous among the other subgroups formed.
- Then, the elements are randomly selected from each of these strata.
- We need to have prior information about the population to create subgroups.

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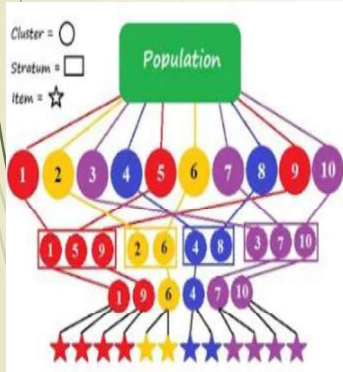
Probability sampling: Cluster sampling (single stage)



- Our entire population is divided into clusters or sections, and then the clusters are randomly selected.
- All the elements of the cluster are used for sampling.
- Clusters are identified using details such as age, sex, location, etc.
- Cluster sampling can be done in the following ways:
- **Single Stage Cluster Sampling:** The entire cluster is selected randomly for sampling.

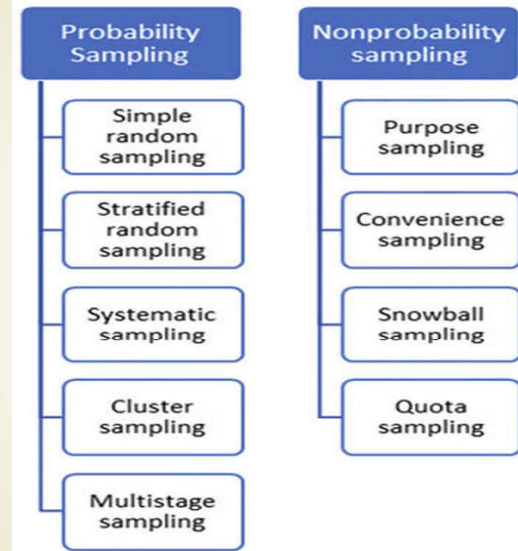
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Multistage Sampling



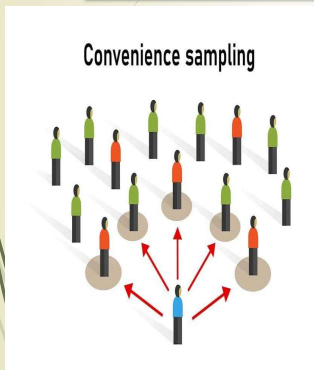
- It is the combination of one or more methods described above.
- Population is divided into multiple clusters and then these clusters are further divided and grouped into various sub groups (strata) based on similarity.
- One or more clusters can be randomly selected from each stratum.
- This process continues until the cluster can't be divided anymore.
- For example country can be divided into states, cities, urban and rural and all the areas with similar characteristics can be merged together to form a strata.

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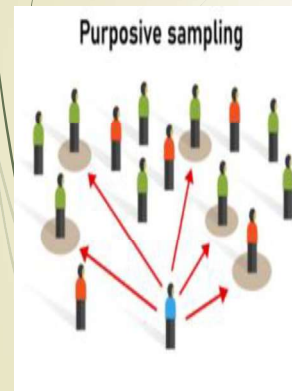
Non-probability sampling: Convenience sampling



- Here the samples are selected based on the availability.
- This method is used when the availability of sample is rare and also costly.
- So based on the convenience samples are selected.
- For example, Researchers prefer this during the initial stages of survey research, as it's quick and easy to deliver results.

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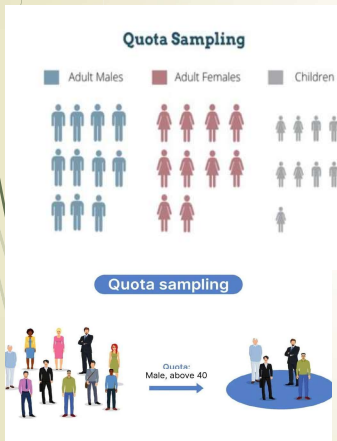
Non-probability sampling: Purposive sampling



- This is based on the intention or the purpose of study.
- Only those elements will be selected from the population which suits the best for the purpose of our study.
- For Example:** If we want to understand the thought process of the people who are interested in pursuing master's degree then the selection criteria would be "Are you interested for Masters in..?"
- All the people who respond with a "No" will be excluded from our sample.

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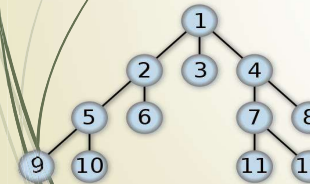
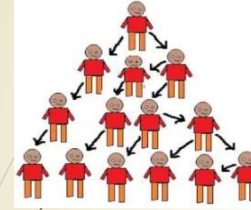
Non-probability sampling: Quota sampling



- ▶ This type of sampling depends on some pre-set standard. It selects the representative sample from the population.
- ▶ Proportion of characteristics/ trait in sample should be same as population. Elements are selected until exact proportions of certain types of data are obtained or sufficient data in different categories is collected.
- ▶ **For example:** If our population has 45% females and 55% males then our sample should reflect the same percentage of males and females.

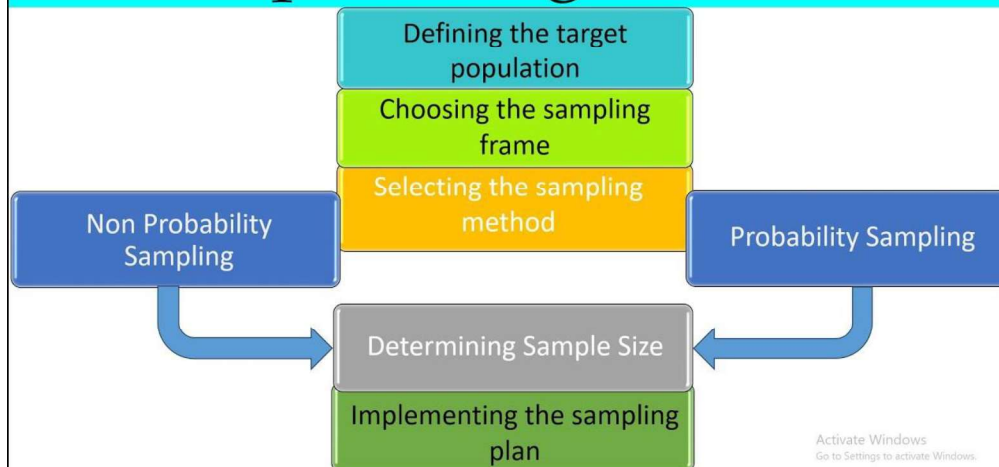
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Non-probability sampling: Referral/Snowball sampling



- ▶ This technique is used in the situations where the population is completely unknown and rare.
- ▶ Therefore we will take the help from the first element which we select for the population and ask him to recommend other elements who will fit the description of the sample needed.
- ▶ So this referral technique goes on, increasing the size of population like a snowball.
- ▶ **For example:** It's used in situations of highly sensitive topics like HIV Aids where people will not openly discuss and participate in surveys to share information about HIV Aids. Not all the victims will respond to the questions asked so researchers can contact people they know or volunteers to get in touch with the victims and collect information. Helps in situations where we do not have the access to sufficient people with the characteristics we are seeking. It starts with finding people to study.

Sample Design Process



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Sampling size

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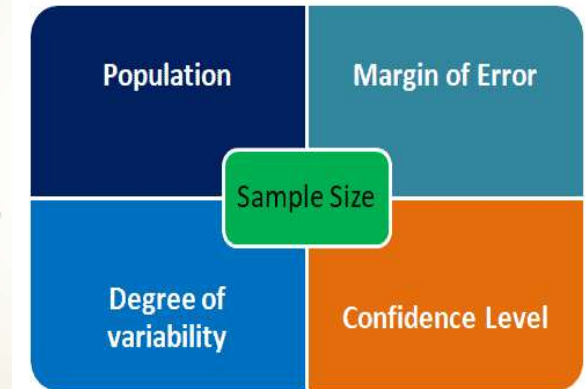


- ❑ Sample size determination is the mathematical estimation of the number of population units considered for the study.
- ❑ In other words, it is the subset of the population selected to make inferences about the population.
 - ❑ It must be adequate to represent the population.

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Factors affecting the sample size?

In a quantitative study, the sample size is selected based on the population, the desired margin of error, the chosen confidence level, and the degree of variability.



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Population and sample size

- Sample size is governed primarily by the desired confidence level, margin of error, and variability in the population, rather than by population size itself.
- Sample size does not increase proportionally with population size.
- Smaller finite populations require smaller samples, whereas for large populations the required sample size approaches a constant value determined mainly by confidence level and precision.

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Margin of error and sample size

- The margin of error is a statistic, expressing the amount of random sampling error in the results of an experiment.
- **Margin of error has a strong inverse relationship with sample size: the smaller the margin of error you want, the larger the sample you need.**
- It determines how much higher or lower the population means you are willing to let your sample mean fall. Often, margin error is **expressed in terms of percentage.**
- **Example:** The error will be higher in case of heterogeneous or diversified population than homogeneous or uniform population. So, what is the permissible limit for the study? Greater precision needs a higher sample size.

Degree of variability of the population and sample size

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In practical terms, if you do not know the expected population variability in a proportion-based study, using $p = 0.5$ is usually recommended because it produces the safest, largest sample-size estimate.

- **Degree of variability:** The degree of variability is the extent to which the sample measures differ from the measure of the population.
- The more heterogeneous or variable the population is, the larger the sample needed to obtain a reliable estimate.
- **The greater the variance, the larger the sample size.**

higher variability → larger sample size,
while lower variability → smaller sample size.

Confidence level and sample size

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95% confidence is the most commonly used level

- **Confidence Level:** The Confidence level is the estimated probability that a population estimate lies within a given margin of error.
- In other words, it tells how confident you want to be that the actual mean falls within your confidence level.
- Higher is the confidence, larger is the sample size, because we want greater certainty that the sample estimate reflects the true population value.
- The most commonly used confidence levels are 90%, 95%, and 99%.

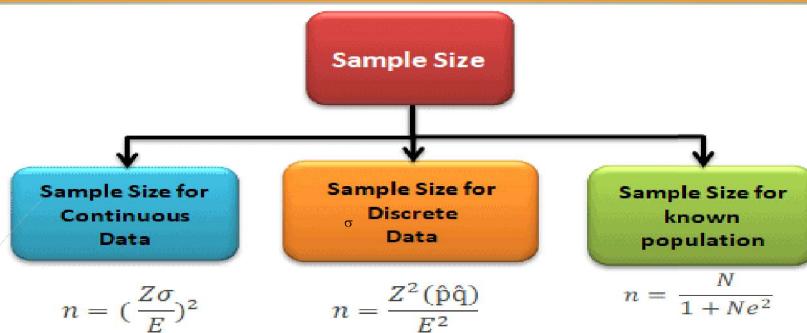
Confidence Level to Z-Score

90%	→	1.645
95%	→	1.96
98%	→	2.33
99%	→	2.575

Higher confidence level → larger sample size
Lower confidence level → smaller sample size

How to calculate a sample size?

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Where n is the sample size, Z is the Z score from the desired risk, sigma is the standard deviation, and E is the mean shift or error.

$p = 1 - q$, Where: n = sample size, p̂ is the variability, q̂ = 1 - p̂. p = population proportion

Where, n = sample size, N = total population, e = margin of error

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Example-1: Sample size for the continuous data

- **Question:** You are given a population with a standard deviation of 8.6. Determine the sample size needed to estimate the mean of the population within ± 0.5 with a 99 percent confidence.
- **Answer:** 99 percent confidence $Z = 2.575$
- where Z is the standard normal ordinate (**need to calculate from the table**)
- Standard value (for 90% confidence it is 1.645; for 95% it is 1.96; for 99% it is 2.575)
- $e = \pm 0.5$
- $\sigma = 8.6$

$$n = \frac{Z^2 \sigma^2}{e^2} = \frac{(2.575)^2 (8.6)^2}{(0.5)^2} = 1961.6 \approx 1962 \text{ (Approximately)}$$

Example-2: Sample size for the discrete data

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- **Question:** A consumer electronics company wants to determine the job satisfaction levels of its employees. For this, they ask a simple question: Are you satisfied with your job? It was estimated that no more than 30 per cent of the employees would answer yes. What should be the sample size for this company to estimate the population proportion to ensure a 95 per cent confidence in result and to be within 0.04 of the true population proportion?
- **Solution:** 95 per cent confidence $Z = 1.96$
- where Z is the standard normal ordinate (need to calculate from table)
- Standard value (for 90% confidence it is 1.645; for 95% it is 1.96; for 99% it is 2.055)
- $e = 0.04$
- $P = 0.3$
- $q = 1 - p = 0.7$
- $n = \frac{Z^2 pq}{e^2} = \frac{(1.96)^2 * 0.3 * 0.7}{0.04^2}$
- $= 504.21 = 505$ (approximately)

Example-3: Sample size for the known population

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- **Question:** Compute the sample size for a population of 1200. You need to take a survey about financial problems. While the margin of error is 5%.
- **Answer:** Population $N = 1200$
- Margin of error $e = 0.05$
- So, Sample size $= N / (1 + Ne^2) = 1200 / (1 + 1200 * 0.05^2) = 1200 / 4 = 300$.

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Class-10: Data Processing

Fieldwork Validation

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- **Fieldwork validation is the process of checking whether the data collected during fieldwork are genuine, properly collected, complete, and in accordance with the research instructions.**
- In simple terms:
 - Check whether the interview really happened to confirm that the fieldworker actually interviewed them.
 - Check the respondent whether the respondent was a real person and belonged to the required target group.
 - Check the fieldworker's work whether the fieldworker followed the instructions given for conducting the survey.
 - Check the questionnaire for missing information, unusual answers, or inconsistencies.
 - Detect false or fabricated data where a fieldworker may have filled in questionnaires without actually conducting interviews.
 - Correct or remove unreliable information

Data editing

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- **Data editing means carefully checking the collected data to find and correct mistakes, missing information, inconsistencies, or unclear responses before the data are analyzed.**
- After collecting questionnaires or responses from respondents, the researcher checks whether the information is:
 - **Complete** – Are all important questions answered?
 - **Accurate** – Are there obvious mistakes in the responses?
 - **Consistent** – Do different answers agree with each other?
 - **Clear** – Is the respondent's answer understandable?
 - **Suitable for analysis** – Can the responses be properly coded and analyzed?
- If problems are found, the researcher may correct them, when possible, clarify them, or exclude the particular response if it cannot be reliably used.

Data editing = checking the collected data and making it ready for analysis.

Continue...

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- Example: Suppose a questionnaire asks:

Age: 25 years
Marital status: Married
Number of children: 15

The answer about 15 children may be checked because it appears unusual or possibly incorrect. The researcher would verify the information rather than directly accepting it.

- Types of editing: Data editing is generally carried out at two stages:

1. **Field editing:** Done soon after the questionnaire is collected, usually by the field supervisor, to identify obvious omissions or errors.
2. **Central editing:** Done later, usually at the research office, where questionnaires are examined more carefully for completeness, consistency, and accuracy.

Coding

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- **Coding means giving numbers or symbols to the answers collected from respondents so that the data can be easily classified, entered, and analyzed.**

- Why is coding needed?

Coding helps to:

1. Organize the data in a systematic way.
2. Reduce lengthy responses into simple numbers or symbols.
3. Make data entry easier.
4. Help classify responses into different categories.
5. Make statistical analysis easier.

Coding = Giving a number or symbol to each answer to make data analysis easier.

Continue....

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- Types of coding in simple terms:

1. **Pre-coding:** Codes are decided before collecting the data. This is common with structured questionnaires where answer options are already provided.
2. **Post-coding:** Codes are decided after collecting the data. This is useful for open-ended questions where respondents give their own answers.

- **Example:** After data editing, the researcher has to organize the responses. For this, each possible answer is given a **code**. For example:

Question	Respondent's answer	Code
Gender	Male	1
Gender	Female	2
Education	Graduate	1
Education	Postgraduate	2
Education	Other	3

Instead of entering the words *Male, Female, Graduate, etc.*, the researcher can enter their corresponding codes.

Classification and tabulation of data

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- Classification and tabulation are steps used to organize the collected data in a meaningful and easy-to-understand form before analysis.
- A simple way to remember them is:
 1. Classification = Grouping the data
 2. Tabulation = Presenting the grouped data in tables
- For example, if a researcher collects information about the age of 100 respondents, the data can be classified as:

Age group	Number of respondents
Below 20	15
20–30	35
31–40	30
Above 40	20

Here, respondents are **classified into different age groups**.

Types of classifications

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Qualitative classification: Data are grouped according to qualities or attributes.

Example:

- Male / Female
- Literate / Illiterate
- Employed / Unemployed

Quantitative classification: Data are grouped according to numerical values.

Example:

Income below ₹20,000
₹20,000–₹40,000
₹40,000–₹60,000
Above ₹60,000

Chronological classification: Data are arranged according to time.

Example:

Year	Sales
2022	₹5 lakh
2023	₹7 lakh
2024	₹9 lakh

Geographical classification: Data are grouped according to place or geographical area.

Example:

Odisha
West Bengal
Bihar
Jharkhand

Continue.....

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- Tabulation means presenting classified data systematically in rows and columns.

Example:

Education level	Number of respondents
School	20
Graduate	45
Postgraduate	25
Other	10
Total	100

This makes the data short, clear, and easy to compare.

Types of Tabulation:

- 1. Simple or one-way table:** Shows information about **one characteristic**.
Example: Number of respondents according to gender.
- 2. Two-way table:** Shows information about **two characteristics**.
Example: Gender and employment status.
- 3. Multiple or manifold table:** Shows information about **more than two characteristics**.

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Why are classification and tabulation important?

They help the researcher to:

- **Organize large amounts of data**
- **Make data easy to understand**
- **Compare different groups**
- **Identify patterns and relationships**
- **Make statistical analysis easier**
- **Present findings clearly**

Statistical Software Packages

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- ▶ Statistical software packages are computer programs used to enter, process, organize, analyze, and interpret research data.
- ▶ In simple words: **Statistical software = Computer programs that help researchers analyze data quickly and accurately.**
- ▶ **What do statistical software packages do?** After data has been edited, coded, classified, and tabulated, statistical software can be used to:
 1. Enter and store data in a computer.
 2. Organize and manage large amounts of data.
 3. Calculate percentages, averages, frequencies, etc.
 4. Prepare tables and charts.
 5. Perform statistical tests.
 6. Find relationships between different variables.
 7. Help researchers interpret the results.

Common statistical software packages:

- ❑ **SPSS:** widely used for social science and survey research.
- ❑ **SAS (System for statistical analysis):** used for advanced statistical analysis and large datasets.
- ❑ **MS Excel:** useful for basic data management, calculations, tables, and charts.
- ❑ **Stata:** commonly used for statistical and econometric analysis.
- ❑ **Minitab:** Developed by Pennsylvania state university

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THANKS